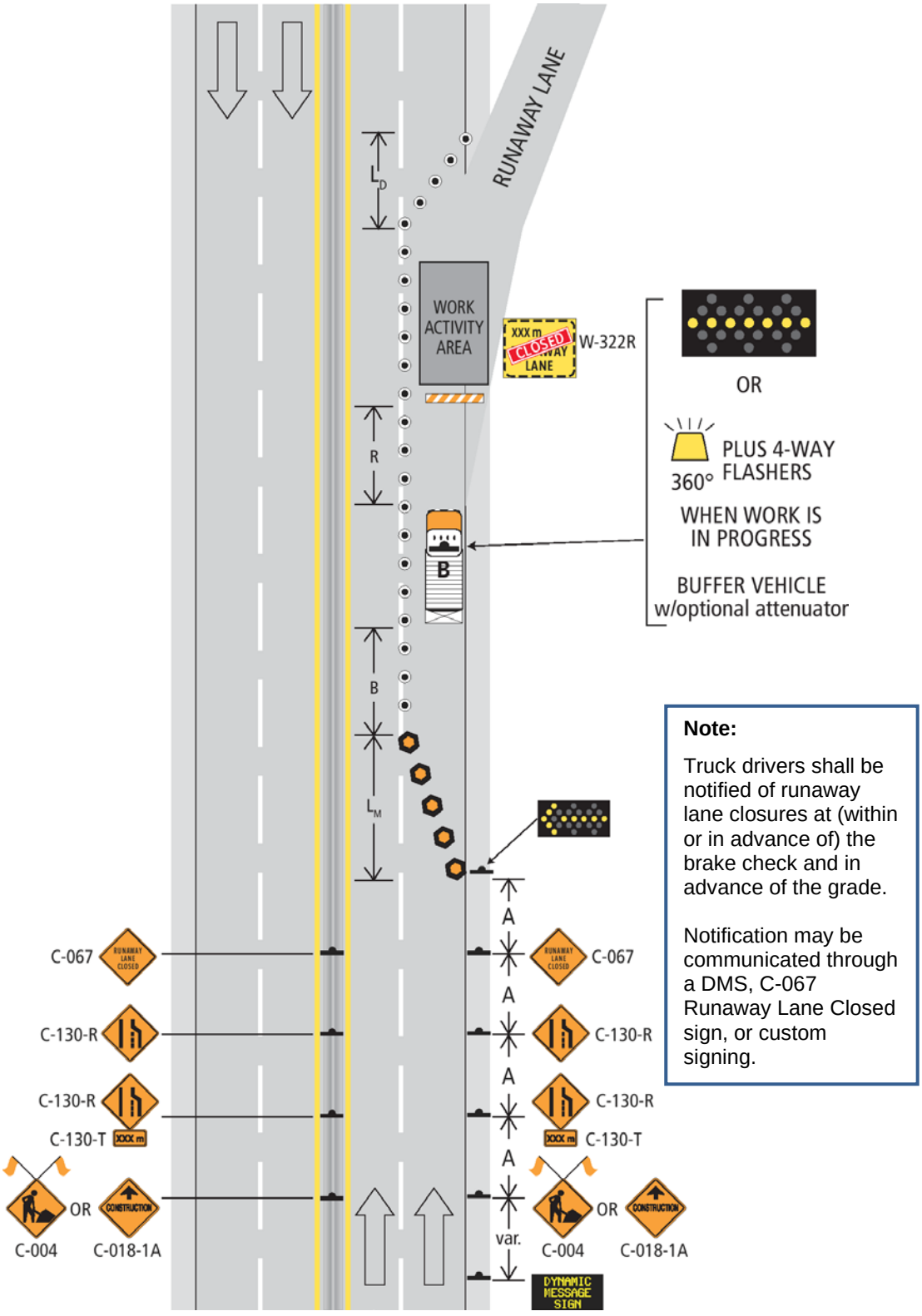
Figure 9.17: Runaway Lane Closed – Short and Long Duration



Note:

Truck drivers shall be notified of runaway lane closures at (within or in advance of) the brake check and in advance of the grade.

Notification may be communicated through a DMS, C-067 Runaway Lane Closed sign, or custom signing.

9.18 Lane Closure with Zipper Merge Signing (Queues < 800 m)

Purpose:

Late merge signs (zipper merge) may be used as an alternate to standard lane closure guidelines in high volume areas where queuing conditions are expected. Late merge signs encourage drivers to utilize both lanes until a merge point which improves efficiency by reducing congestion and aggressive driving. Late merge signs should not be used within intersections or on sharp horizontal or vertical curves. This layout shows the typical setup of late merge signs for the closure of a left travel lane on a multilane undivided highway.

For locations where queueing may extend more than 800 m upstream of the taper, refer to *Section 9.19: Lane Closure with Zipper Merge Signing (Queues ≥ 800 m)*.

Standard:

- A Lane Closed Ahead C-130 sign with a C-130-T distance tab is required on both sides of the roadway
- A barricade is required on the approach side of the work activity for long-duration work

Guidance:

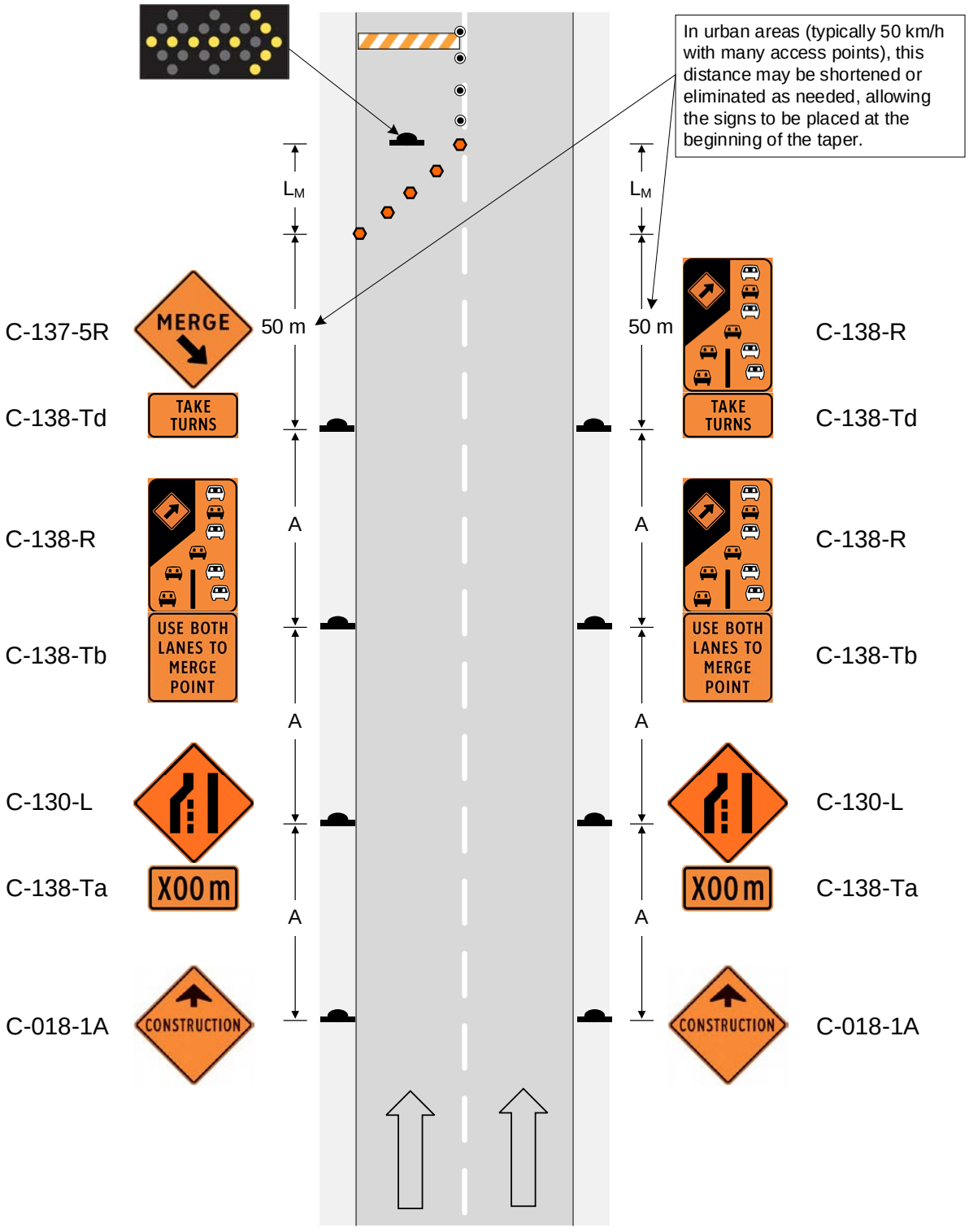
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*

Options:

- Additional confirmatory Zipper Merge C-138-LR signs with accompanying C-138-Ta distance tabs may be used with a maximum 2km spacing where queuing extends beyond the C-130 signs or as considered necessary
- If the regular posted speed limit is ≤ 60 km/h, tubular markers may be used for leading tapers instead of drums.
- Dynamic message sign (DMS) may be used in place of the first set of C-138-LR and accompanying C-138-Tb (on both sides of roadway) but should include the messaging "Zipper Merge. Use Both Lanes to Merge Point," or similar
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.18: Lane Closure with Zipper Merge Signing (Queues < 800 m)



9.19 Lane Closure with Zipper Merge Signing (Queues ≥ 800 m)

Purpose:

The late merge (zipper merge) sign layouts are alternatives to standard lane closure set-ups on multilane divided highways. The goal of the zipper merge sign layout is to achieve better lane utilization and merging behavior to minimize delays and lengths of queues.

Late merge signs encourage drivers to utilize both lanes until a merge point. The signs also seek to improve merge behaviour (take turns) which will improve efficiency and safety by reducing congestion and aggressive driving. Late merge signs should not be used within intersections or on sharp horizontal or vertical curves.

This layout shows the typical late merge signage for the closure of a left travel lane on a multilane divided highway. This layout may be used where the regular posted speed limit is ≥ 70 km/h and traffic queues may extend ≥ 800 m upstream of the taper.

Standard:

- Signs shall be placed on both the left and right side of the roadway.
- A barricade is required on the approach side of the work activity for long-duration work

Guidance:

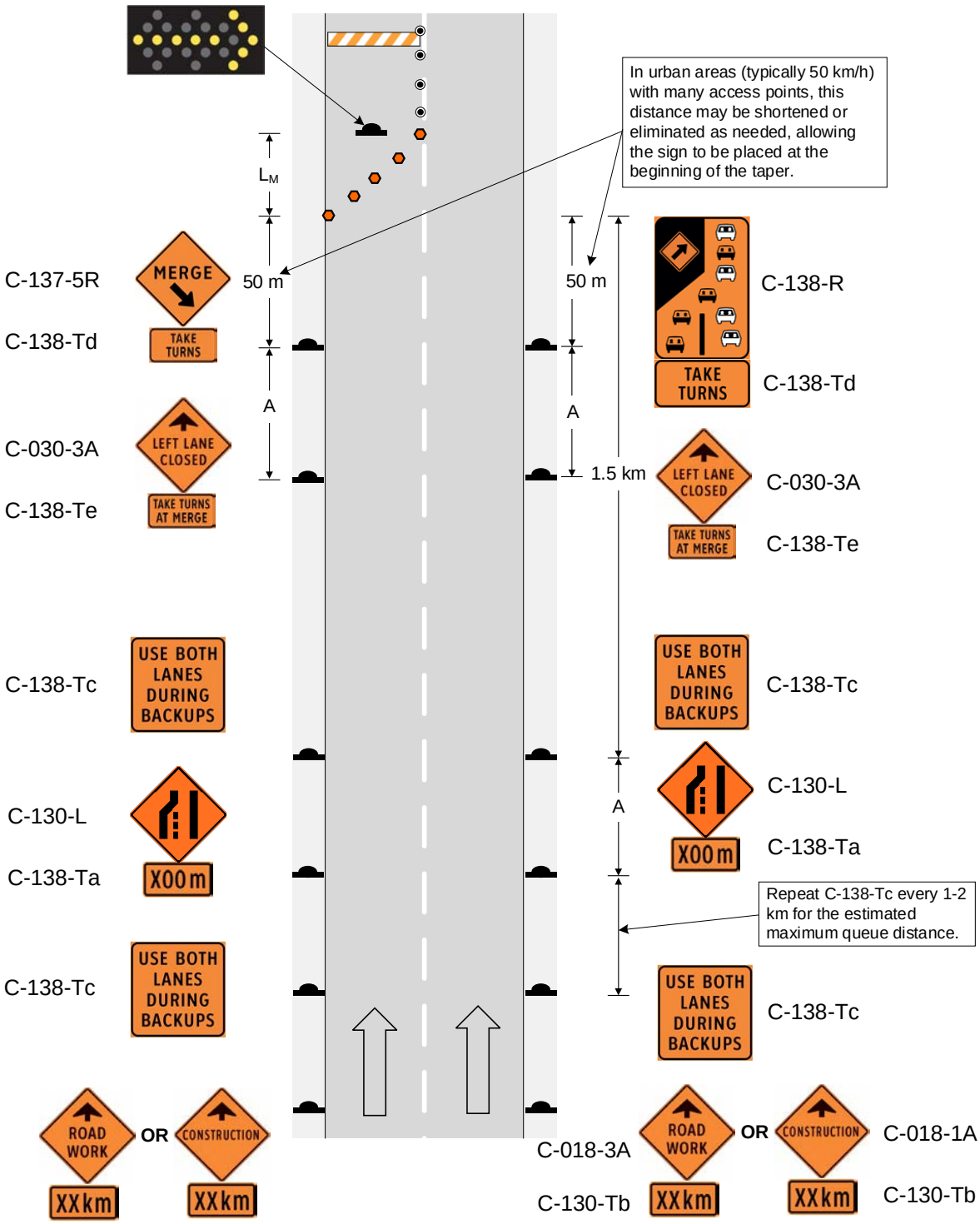
- The Road Work Ahead C-018-3A or Construction Ahead C-018-1A signs should be placed outside the location of where the maximum queue distance is anticipated to develop.
- Distance advisory tabs should be utilized with the Road Work Ahead C-018-3A or Construction Ahead C-018-1A signs to inform drivers where the merge location is located.
- For queue lengths greater than 1.5 km, adding additional Use Both Lanes During Backups C-138-Tc signs improves driver compliance.
- Where shoulders are wide and may be mistaken for travel lanes, shoulder tapers should be considered and installed in accordance with *Table A – Taper Lengths*.

Options:

- Additional confirmatory Zipper Merge C-138-LR signs with accompanying C-138-Ta distance tabs may be used with a maximum 2km spacing where queuing extends beyond the C-130 signs or as considered necessary
- If the regular posted speed limit is ≤ 60 km/h, tubular markers may be used for leading tapers instead of drums.
- A dynamic message sign (DMS) with similar messaging may be used in place of a pair of USE BOTH LANES DURING BACKUPS C-136-T7 signs
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 9.19: Lane Closure with Zipper Merge Signing (Queues ≥ 800 m)



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PART B – TRAFFIC CONTROL

Section 10: Traffic Control Layouts – Mobile Work

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PART B – TRAFFIC CONTROL

Section 10: Traffic Control Layouts – Mobile Work

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

10.1 General Information - Mobile Work

Mobile work is an operation that is either continuously slow-moving work or intermittently-moving work with short stops. It may include:

- shouldering
- gravel road grading
- multiple pothole patching
- mowing
- brushing
- sweeping
- hydro-seeding
- sign cleaning
- litter bag pick-up
- spraying for dust control
- concrete barrier scupper flushing

Mobile work does not include debris removal or emergency response.

Traffic control devices are typically vehicle-mounted and include Slow Vehicle Next X km C-044 signs with the specified distance not exceeding 8 kilometres (the distance between the two C-044 signs in opposing directions) or Truck Stopped on Road Next 2 km C-038 signs.

If an advance Slow Vehicle Next X km C-044 sign is not used, a shadow vehicle with vehicle-mounted traffic control devices is required.

The distance between the work and shadow vehicles is typically Distance A found in Table B. It may vary on the basis of site factors such as traffic volume, sight distance, and terrain.

Table D shall be used to determine whether or not the operation is mobile work.[1]

TABLE D – MINIMUM DISTANCES FOR MOBILE WORK

	Regulatory Speed Limit (km/h)					
	≤50	60	70 - 80	90	100	≥ 110
Minimum Distance Moved (m) Every 30 Minutes	100	130	170	220	260	300

Table D Notes

To be considered mobile work, the operation shall move at least the distance shown in Table D for the posted speed limit every 30 minutes or less.

If the work does not regularly move the specified distance, it should be treated as a stationary operation, and the appropriate layout should be used.

[1] Table D values are derived from the Minnesota Department of Transportation's Temporary Traffic Control Manual (2011).

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- All work and shadow vehicles shall be equipped with a flashing arrow board (FAB) in caution mode, a 360 degree flashing light, and 4-way flashers.
- Work and shadow vehicles shall display rear-mounted Slow Moving Vehicle C-036 signs.

Guidance:

- All temporary signs should be removed or covered when work is not under way.
- When the regular posted speed limit is ≥ 70 km/h, a rear-mounted crash attenuator is recommended for the shadow vehicles.

Options:

- A portable dynamic message sign (DMS) may be used to provide advance messaging for drivers.

10.2 Continuously Slow-Moving Work – Two-Lane, Two-Way Roadway

Purpose:

Continuously slow-moving work is done while continuously moving at slow speeds. These operations normally do not involve stopping, but infrequent stops of up to 15 minutes may occur.

Examples include:

- shouldering
- mowing
- sweeping
- concrete barrier scupper flushing
- gravel road grading
- spraying for dust control
- hydro-seeding
- brushing

For pavement marking layouts (i.e., intersection marking and quick-dry pavement marking), see *Section 14: Traffic Control Layouts – Pavement Marking.*

Standard:

- Work and shadow vehicles shall display Yield to Oncoming Traffic R-56-1 signs.
- Slow Vehicle Next X km C-044 signs shall be used, and they shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the signs.
- If the speed limit is ≥ 70 km/h, a shadow vehicle displaying the Slow Vehicle(s) Ahead C-045-X sign is required.

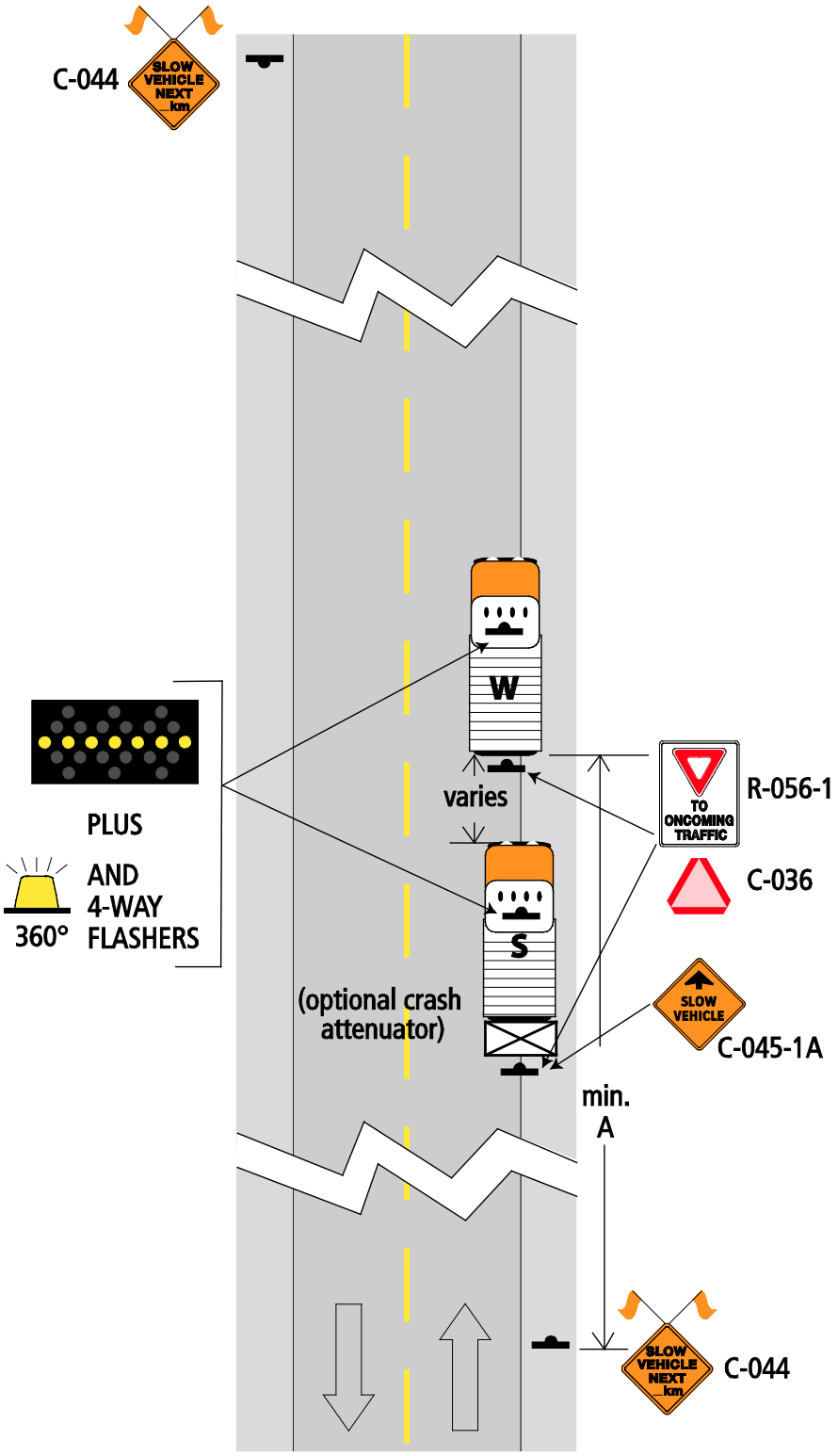
Guidance:

- The distance on Slow Vehicle Next X km C-044 signs should not exceed 8 kilometres.
- Work and shadow vehicles should pull over periodically to allow queued traffic to pass.
- The shadow vehicle should keep as far right as practicable, but may encroach into the travel lane when the shoulder is too narrow to drive on.
- Work vehicles may travel at posted speeds when work is not under way.

Options:

- The shadow vehicle may be omitted for work on low volume roadways (<1000 vpd) with a speed limit ≤ 80 km/h. These are typically non-numbered highways and may include rural highways with statutory speed limits of 80 km/h and appropriate sight distances.
- The Slow Vehicle(s) Ahead C-045-X sign may be replaced with other appropriate signs related to the type of work (see *Appendix B: Standard Construction Signs*).
- If the speed limit is ≤ 60 km/h:
  - For low-volume roadways, the Slow Vehicle Next X km C-044 sign and/or shadow vehicle and its accessories may be omitted.
  - The flashing arrow board (FAB) on the work vehicle may be omitted but the 360-degree flashing light and 4-way flashers are always required.
- The Yield to Oncoming Traffic R-056-1 sign may be omitted from large, line-type utility vehicles if it is impractical to mount the sign.

Figure 10.2: Continuously Slow-Moving Work – Two-Lane, Two-Way Roadway



10.3 Intermittently-Moving Work – Two-Lane, Two-Way Roadway

Purpose:

Intermittently-moving work is a maintenance activity for which the work zone changes frequently or a work operation that involves frequent short stops not exceeding 30 minutes in duration.

Examples include:

- group relamping of street lights
- multiple pothole patching (several stops within 1 km)
- traffic control device placement and/or removal
- crack sealing
- sign cleaning
- catch basin flushing
- station measurement
- litter bag pick-up
- raised pavement marker installation

Standard:

- Work and shadow vehicles shall display Yield to Oncoming Traffic R-56-1 signs.
- The shadow vehicle shall display a rear-mounted Caution This Truck Stops Frequently C-039 sign.
- A Truck Stopped on Road Next 2 km C-038 sign is required, and it shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the sign.

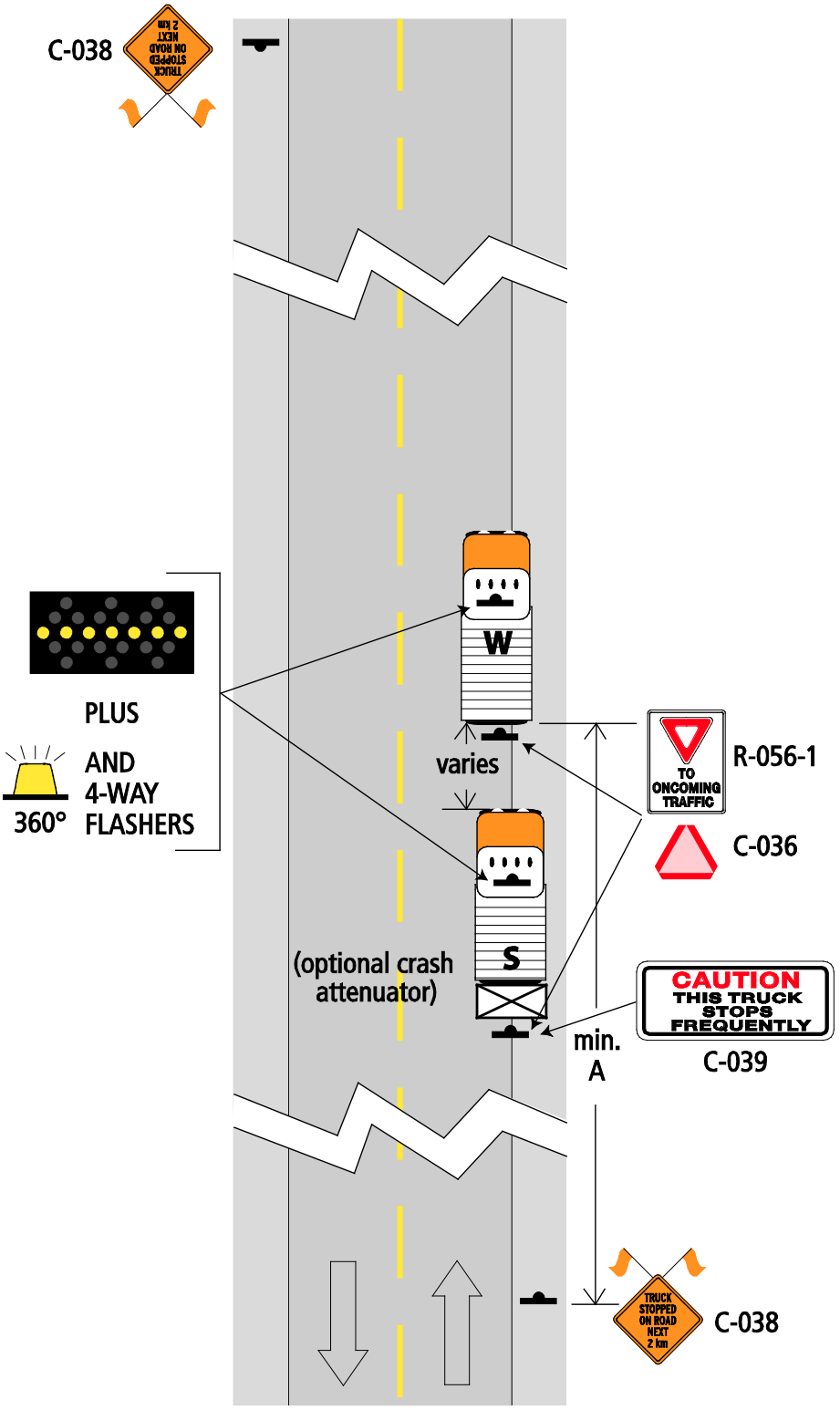
Guidance:

- The shadow vehicle should keep as far right as practicable—or as far left as practicable if working in the left lane—but may encroach into the travel lane when the shoulder is too narrow to drive on.
- Work vehicles may resume travelling at posted speeds when work stops.

Options:

- A shadow vehicle may not be required for work that typically occurs off the roadway on the shoulder (e.g., litter bag pick-up or sign-cleaning activities).
- If the construction speed limit is ≤ 60 km/h:
  - Where an advance Truck Stopped on Road Next 2 km C-038 sign is used, using a shadow vehicle is optional.
  - For low-volume roadways, the Truck Stopped on Road Next 2 km C-038 sign and/or a shadow vehicle and its accessories may be omitted.
  - The flashing arrow board (FAB) on the work vehicle may be omitted, but the double-sided Prepare to Stop C-040D sign must be used in its place. The 360-degree light and 4-way flashers are always required
- An arrow stick or a shadow vehicle may be substituted for the flashing arrow board (FAB) on large, line-type utility vehicles if it is impractical to mount the FAB.

Figure 10.3: Intermittently-Moving Work – Two-Lane, Two-Way Roadway



10.4 Continuously Slow-Moving Work – Multilane Undivided or Divided Roadway

Purpose:

Continuously slow-moving work is done while continuously moving at slow speeds. These operations normally do not involve stopping, but infrequent stops of up to 15 minutes may occur.

Examples include:

- grading/shouldering
- mowing
- concrete barrier flushing
- dust control
- hydro-seeding
- sweeping

For pavement marking layouts (i.e., intersection marking and quick-dry pavement marking), see *Section 14: Traffic Control Layouts – Pavement Marking.*

Standard:

- Shadow vehicles shall display a Slow Vehicle(s) Ahead C-045-X sign.
- At least one shadow vehicle shall be used for undivided roadways.
- Two shadow vehicles shall be used for divided roadways.
- Slow Vehicle Next X km C-044 signs shall be used, and they shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the signs.

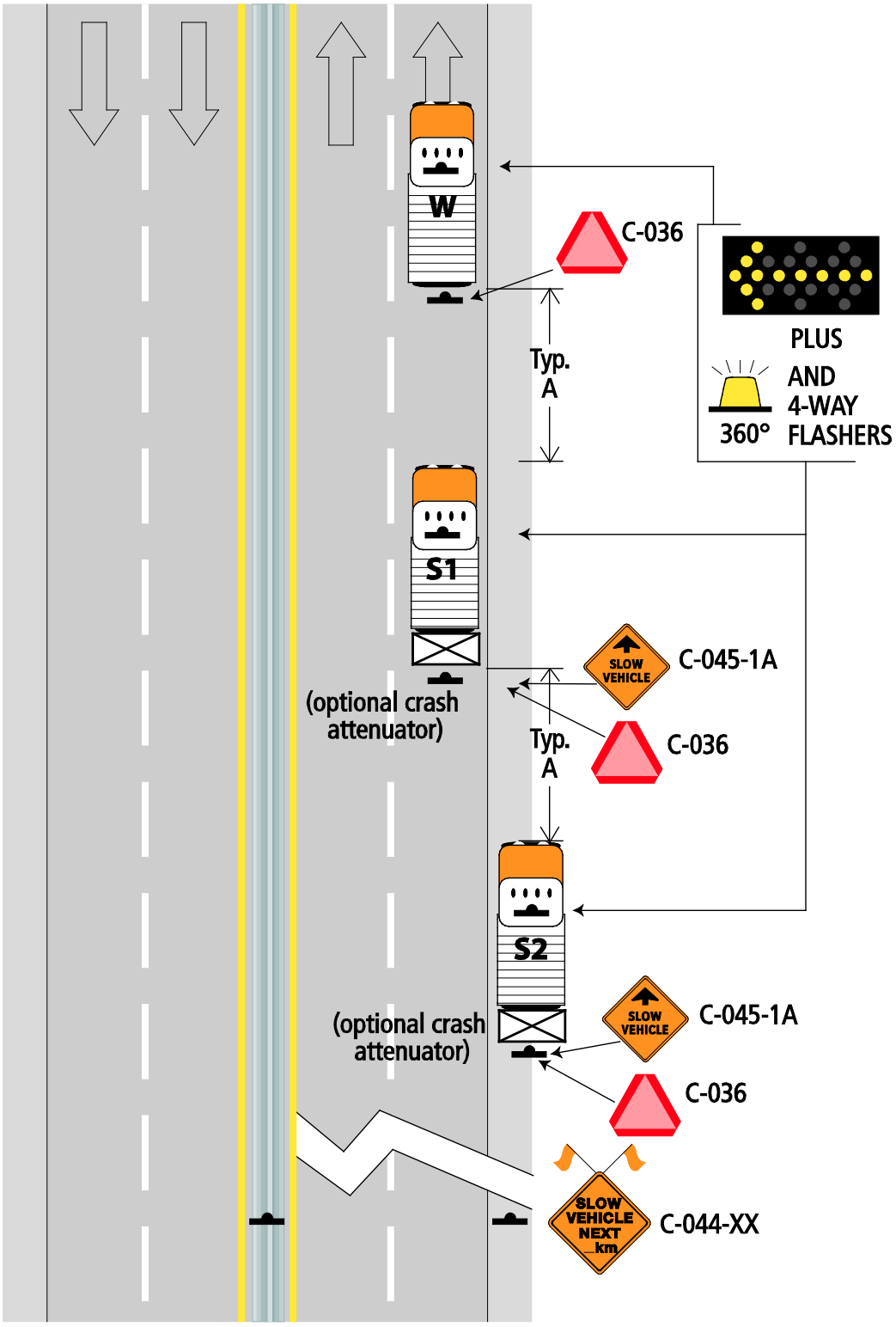
Guidance:

- The distance on the Slow Vehicle Next X km C-044 signs should not exceed 8 kilometres.
- Typical spacing between the shadow vehicle and the work vehicle is Table B Distance A.
- For divided roadways, the shadow vehicle should keep as far right as practicable—or as far left as practicable if working in the left lane—but may encroach into the travel lane when the shoulder is too narrow to drive on.
- The space between the shadow vehicle and the work vehicle may be adjusted as follows:
  - decreased in areas with heavy traffic to deter road users from driving between the two vehicles
  - increased to provide adequate sight distance for vehicles approaching from the rear

Options:

- If the speed limit is ≤ 60 km/h:
  - Where a Slow Vehicle Next X km C-044 sign is used, using one less shadow vehicle is optional.
  - The flashing arrow board (FAB) on the work vehicle may be omitted but the 360-degree flashing light and 4-way flashers are always required.

Figure 10.4: Continuously Slow-Moving Work – Multilane Undivided or Divided Roadway



10.5 Intermittently-Moving Work – Multilane Undivided or Divided Roadway

Purpose:

Intermittently-moving work is a maintenance activity for which the work zone changes frequently or a work operation that involves frequent short stops not exceeding 30 minutes in duration.

Examples include:

- group relamping of street lights
- multiple pothole patching (several stops within 1 km)
- traffic control device placement and/or removal
- crack sealing
- sign cleaning
- catch basin flushing
- station measurement
- litter bag pick-up
- pavement marker installation

Standard:

- Work vehicles shall display rear-mounted Slow Moving Vehicle C-036 signs and Caution This Truck Stops Frequently C-039 signs.
- Shadow vehicle shall display rear-mounted Slow Moving Vehicle C-036 signs and Slow Vehicle(s) Ahead C-045-X signs.
- At least one shadow vehicle shall be used for undivided roadways.
- Two shadow vehicles shall be used for divided roadways.
- A Truck Stopped on Road Next 2 km C-038 sign is required, and it shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the sign.

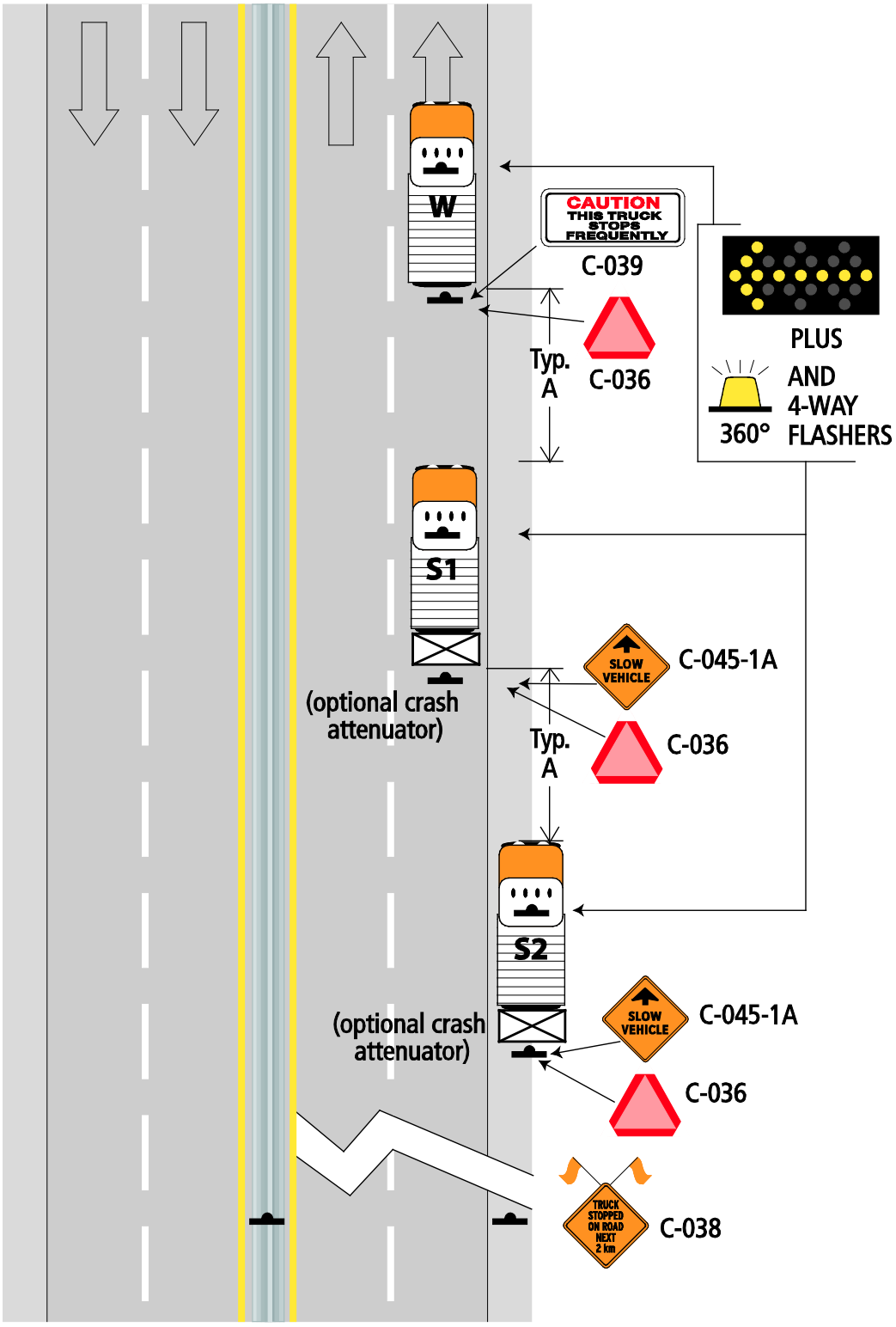
Guidance:

- For divided roadways, the shadow vehicle should keep as far right as practicable—or as far left as practicable if working in the left lane—but may encroach into the travel lane when the shoulder is too narrow to drive on.
- Typical spacing between the shadow vehicle and the work vehicle is Table B Distance A.
- The space between the shadow vehicle and the work vehicle may be adjusted as follows:
  - decreased in areas with heavy traffic to deter road users from driving between the two vehicles
  - increased to provide adequate sight distance for vehicles approaching from the rear

Options:

- If the speed limit is ≤ 60 km/h and if two shadow vehicles are used:
  - Where a Truck Stopped on Road Next 2 km C-038 sign is used, using one less shadow vehicle is optional.
  - The flashing arrow board (FAB) on the work vehicle may be omitted but the 360-degree flashing light and 4-way flashers are always required.

Figure 10.5: Intermittently-Moving Work – Multilane Undivided or Divided Roadway



10.6 Rolling Slowdown

Purpose:

A rolling slowdown uses shadow vehicles to protect moving work in the travel lanes, and can be implemented only on a multilane roadway on which passing over the centreline is not permitted.

The shadow vehicles form a moving blockade across all lanes, thereby reducing traffic speeds and creating a large gap in traffic—a clear area in which work can be accomplished without completely stopping the traffic.

This setup is typically used in circumstance where a full road closure would otherwise be required for short-duration work operations, and for which implementing traffic control measures would take more time than the actual work (e.g., a film company taking a scenic shot along a roadway).

Using this type of setup is resource-intensive and requires consultation with the Road Authority before it can be initiated.

Standard:

- All ramps and entrances to the roadway between the moving blockade and the work operation shall be temporarily closed using a shadow vehicle.
- Each of those ramps shall remain closed until the crew doing the work gives the “all clear” signal or until the front of the moving blockade passes the closed on-ramps and entrances.
- At least one shadow vehicle shall be used in each lane,
- The shadow vehicles blocking traffic shall enter the roadway far enough upstream from the work area to allow a clear area to develop in front of them.
- The shadow vehicles shall move into position so that they form a moving blockade across the travel lanes.
- A separate chase vehicle with a 360-degree flashing light and 4-way flashers shall follow the last public vehicle ahead of the blockade, travelling in front of the moving blockade of shadow vehicles.
- As the blockade slows down, it creates the work area between the blockade and the chase vehicle.
- Communications shall be maintained between the work crew and the moving blockade so that the speed of the blockade can be adjusted to increase or decrease the closure time if necessary.
- Traffic can be released only after all workers and their vehicles have been confirmed to be clear of the roadway.

To calculate the distance upstream of the work area at which to start the slowdown:

A long enough gap in traffic shall be created for traffic moving at a fixed, reduced speed to provide the estimated time needed for the work to be done.

Rolling Slowdown Calculations:

Known:

T = Time needed with no traffic (in minutes).
Vs = Speed of slowdown vehicles (in km/h). A minimum 30 km/h speed limit is recommended.
Vc = Speed of chase vehicle in front of slowdown (in km/h). It should generally be the posted speed limit.

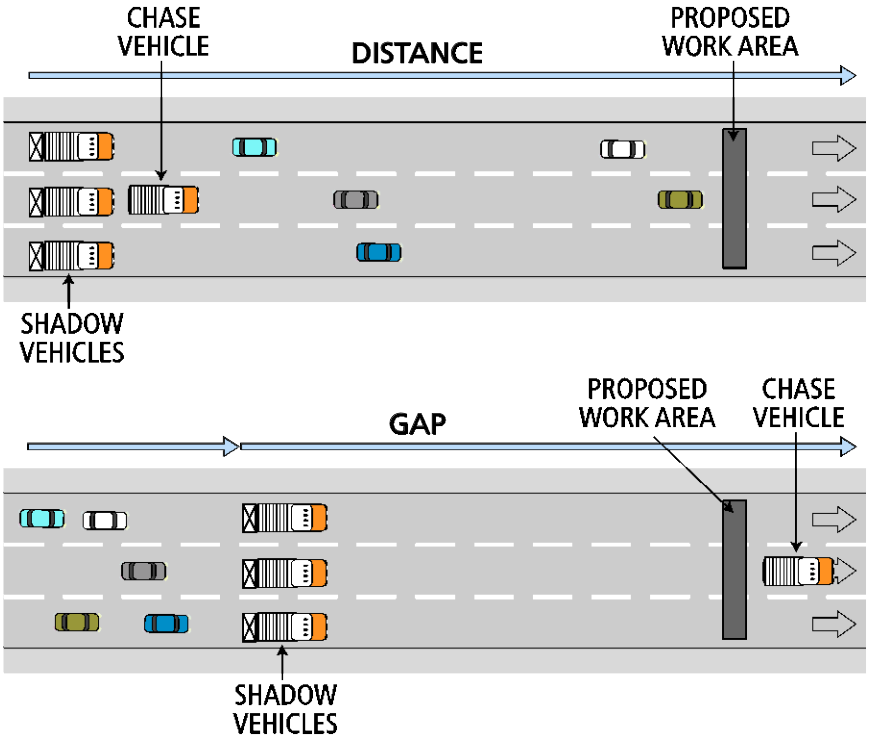
Calculations:

G = Gap needed (in km).
G = T (Vs/60)

C = Clearance time needed to create the required gap (in minutes).
C = G / (Vc/60 - Vs/60)

D = Distance ahead of the work area at which to start the slowdown (in km).

Example:



continued →

Calculations (continued):

A 5-minute gap is required on a 100 km/h freeway to move a large piece of equipment across the roadway and into the median work area, so a 30 km/h rolling slowdown is proposed during the off-peak or lowest traffic volume hours for the freeway.

G = 5 (30/60) = 2.5 km

C = 2.5 / (100/60 - 30/60) = 2.9 minutes

D = Distance ahead of the work area at which to start the slowdown (in km).

D = 2.9 (100/60) = 4.9 km

Guidance:

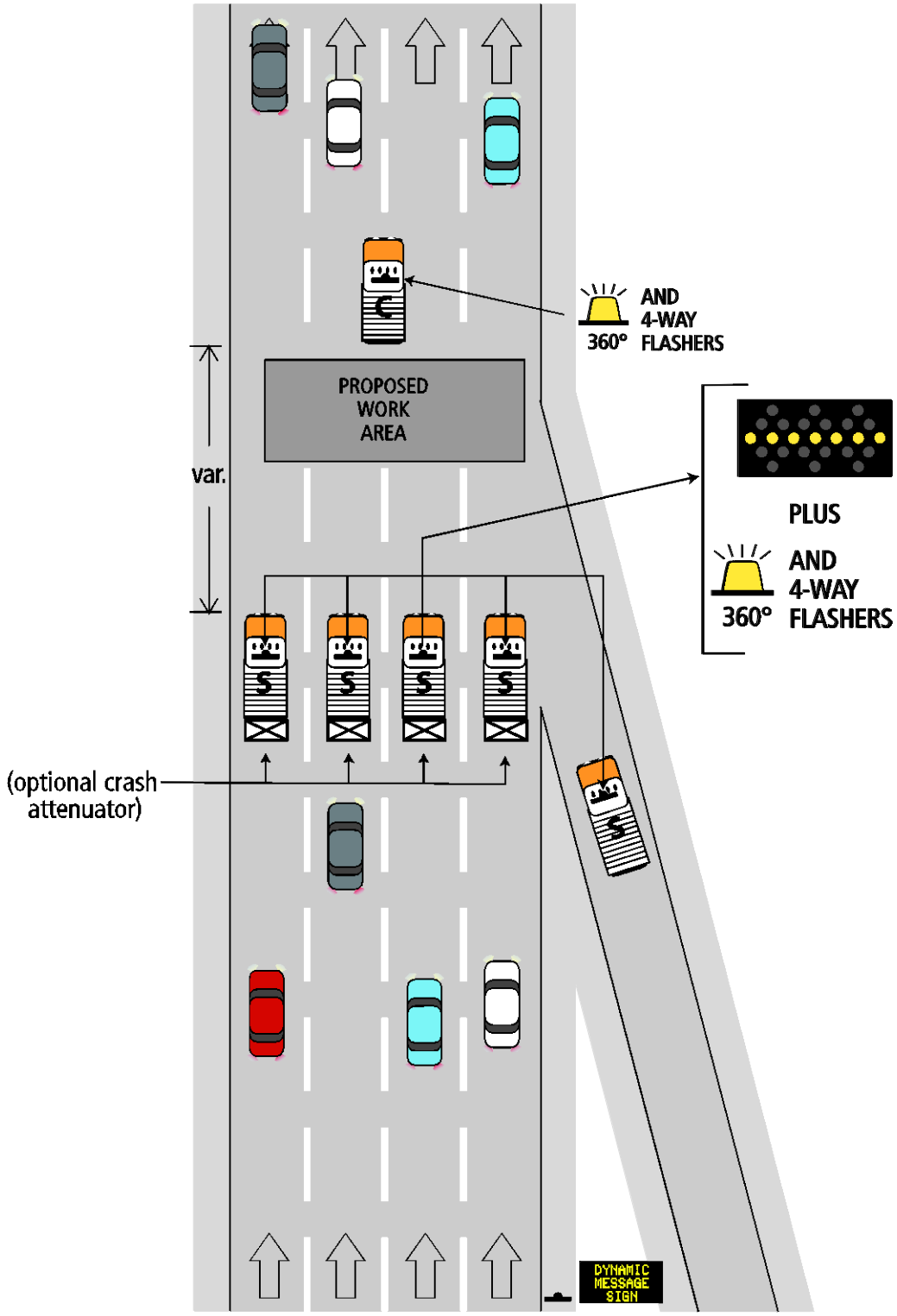
- The shadow vehicles in the rolling slowdown should travel on the lane lines.

Options:

- A truck-mounted dynamic message sign (DMS) may be used to display this message:

Slow or Stopped Vehicles Ahead

Figure 10.6: Rolling Slowdown



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PART B – TRAFFIC CONTROL

Section 11: Traffic Control Layouts – Intersections[1]

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1 See *Section 12: Traffic Control Layouts – Roundabouts* for information on traffic control layouts for roundabouts.